

MOHAMED ATOUI

Machine Learning Engineer · Researcher · Builder

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SUMMARY

Final-year BSc Computer Science & AI student at Royal Holloway (First-Class), graduating Summer 2026. I work across the research-to-production stack: published academic research on adversarial robustness; built diffusion models from first principles; led KV-cache research that cut Llama-3.1-8B inference memory by 62%; co-founded Nordlys Labs and shipped Hypernova, a Mixture of Models framework that reached #1 on SWE-bench at 75.6% serving 500+ users. Comfortable moving between research, engineering, and customer-facing work in fast-moving environments.

EXPERIENCE

Co-founder & ML Research Lead, [Nordlys Labs](#)

Feb 2025 – Present

- Built **Hypernova**, a Mixture of Models framework that lets developers compose their own LLM routers across heterogeneous backends. Reached **#1 on SWE-bench at 75.6%** and served 500+ users in production.
- Designed an **intelligent routing layer** that scores prompt complexity and dispatches inference across 30+ models, reducing inference cost 50–60% versus running frontier models directly.
- Built **evaluation and observability pipelines** tracking accuracy, retrieval quality, latency, and per-model cost in real time. Defined corrective-action workflows when production performance regressed.
- Worked directly with non-technical clients to gather requirements and translate technical work into commercial-grade product surfaces, real-time dashboards, and integration guides.

Machine Learning Researcher, [Information Security Group \(ISG\), Royal Holloway](#)

Dec 2024 – Sep 2025

- Researched **adversarial robustness and representation learning** in ML-based perception and intrusion-detection systems for autonomous vehicles, analysing failure modes under attack and distribution shift.
- Designed and trained **CNN and LSTM models** in PyTorch; built an end-to-end evaluation framework with controlled ablations on real-world automotive sensor datasets.
- Co-authored **academic reports and technical whitepapers** documenting methodology, experimental setup, and findings for technical reviewers.

Product Data Intern, [Stats Perform](#)

Jun 2025 – Oct 2025

- Analysed large-scale sports datasets (NFL, NBA, MLB) using SQL and Python; built feature-engineering pipelines and feature-selection algorithms for production analytics products.
- Modelled revenue projections for AWS marketplace distribution using **Monte-Carlo simulations**, informing go-to-market strategy.
- Worked directly with **enterprise clients** to scope analytics deliverables and presented findings to cross-functional product teams.

Teaching Assistant, [Royal Holloway, University of London](#)

Oct 2025 – Present

- Mentor **50+ students** in Software Engineering and Mathematical Methods; distil complex technical concepts into accessible learning materials and provide structured written feedback on assignments.

SELECTED PROJECTS

[StratigraphicKV, Hybrid KV-Cache Method \(Undergraduate FYP\)](#) · Python · PyTorch · HuggingFace Transformers

- Benchmarked four attention mechanisms (**MHA, MQA, GQA, MLA**) in iso-parameter 16M-param transformers under controlled experimental conditions; MLA reached **13–26% lower perplexity** than MHA at 72% smaller cache.
- Built **StratigraphicKV**, a layered method combining approaches across the KV-cache literature: classifies attention heads as retrieval or streaming and swaps the streaming cache for O(1) recurrent state. On Llama-3.1-8B: **62% memory reduction at +0.09 perplexity**. Self-directed, topic and direction my own.

[ZeroToDiffusion, Diffusion Models From First Principles](#) · Python · PyTorch

- From-scratch implementation of **DDPM and a VAE** in PyTorch. Derived the variational lower bound, implemented forward and reverse processes, tested each component (noise schedule, loss, sampler) in isolation before composition.
- Built without library wrappers to develop a working understanding of objective decomposition, inductive biases, and reverse-process parameterisation.

[Vantage, Agentic AI Assessment Platform](#) · Python · LangGraph · ChromaDB · Sentence Transformers · FastAPI

- Built an **agentic system orchestrated with LangGraph** executing multi-step reasoning across feasibility scoring, three-year financial projections, and build-vs-buy analysis. Compresses 40+ consultant hours into a 5–10 minute evaluation.

- Implemented a **RAG pipeline with Sentence Transformers and ChromaDB** over 200+ real-world case studies; tracked retrieval hit rate and recommendation accuracy as core product metrics.

CrossGen, Multi-Stage LLM Reasoning Pipeline · Python · FastAPI · Claude API · Pydantic

- Designed a **6-stage agentic reasoning pipeline** (Gentner's Structure-Mapping Theory) that decomposes problems into domain-neutral relations and discovers solutions by mapping structural parallels across 42 curated domains.
- Hybrid deterministic + LLM architecture with **mandatory weakness documentation and falsifiable predictions** for every output, built for evaluability and reliable production behaviour.

TurboQuant, Vector Quantisation Library (co-built with a friend) · Zig · SIMD · Python

- Co-built a Zig implementation of Google's **TurboQuant** online vector-quantisation algorithm: **~3 bits/dim** ($\approx 6\times$ compression) of float embeddings with near-perfect recall@10. Targets memory-efficient inference under tight resource budgets.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, Java, C, SQL

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, NumPy, Pandas, ONNX

Generative & Representation Learning: Diffusion (DDPM), VAEs, autoregressive transformers, attention mechanisms (MHA/MQA/GQA/MLA), self-supervised methods

Agentic AI & LLMs: LangGraph, LangChain, Claude API, RAG, Sentence Transformers, ChromaDB, multi-agent orchestration, prompt engineering

Research Methods: Ablation studies, systematic benchmarking, adversarial robustness, hypothesis-driven experimentation, technical writing

Infrastructure: Linux, Docker, Git, GitLab CI/CD, FastAPI, AWS, Azure, GPU training

Languages: English (fluent) · French (fluent) · Arabic (native)

EDUCATION

Royal Holloway, University of London

Sep 2023 – Jun 2026

BSc (Hons) Computer Science & Artificial Intelligence · **First-Class Honours**

Quantum Computing (projected 90%+, Term 1 2025–26) · **Computational Finance**, Machine Learning, Deep Learning, Natural Language Processing, Statistics.

HONORS & AWARDS

- **International Youth Robotics Competition** (2020): placed 1st; the team was honoured by the President of Algeria. I was 14.
- **E-commerce business** (2024): co-founded with two friends after my first year of university; borrowed £1,500 each, ran sourcing/marketing/fulfilment, and turned a profit within three months. Business is still operating today.
- **MoM Framework public talk** (Feb 2026): presented Hypernova's Mixture of Models architecture to a technical audience.